

OPPORTUNITIES

USER PERSONAS

HOW MIGHT WE HELP PEOPLE ORGANIZE AND RETRIEVE THEIR NOTES MORE EFFECTIVELY?

PROBLEM

Many users spend too much time searching for old information because their notes are stored in different places.

Handwritten notes, screenshots, and scanned documents are difficult to search, while typing is inconvenient when users are busy.

Users may also forget deadlines, lose information, or be unable to access their notes without an Internet connection. In addition, some note-taking applications are too complicated for casual users.



SOLUTION

Opportunities—A Multifunction Note-Taking App

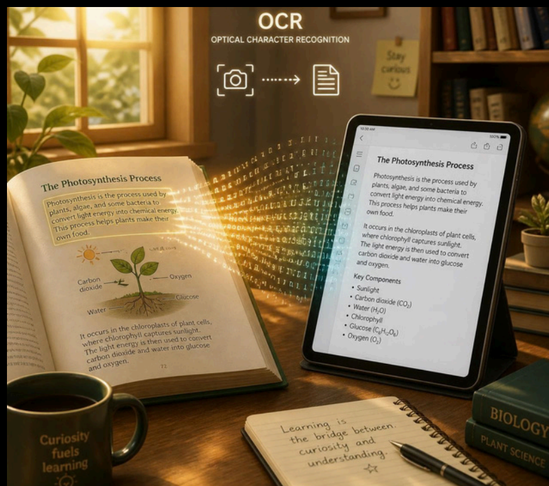
Provides one platform for creating notes through text, voice, photos, scans, drawings, checklists, and tables.

Users can organize their notes with folders, tags, colors, pins, and links. The application also supports OCR search, reminders, automatic saving, offline access, synchronization between mobile and web, and interface customization through themes, font sizes, and focus mode.









SYMPATHY MAP

THINK & FEEL

- "I have too many notes, but I cannot remember where I saved them."
- "I want to write something down quickly before I forget it."
- "I need to access my notes on both my phone and my computer."
- "I am afraid of losing important information."
- "I want the application to be simple and not confusing."
- "I need reminders for important tasks and deadlines."

HEAR

- Friends recommend using different applications for notes, tasks, documents, and reminders.
- Teachers and colleagues remind them to record important information.
- People often say that they forget tasks because they do not write them down.
- Users hear about cloud synchronization but worry about privacy and Internet dependency.
- Some people recommend using paper notebooks because they are easier to use.

PAIN

- Important notes are difficult to find.
- Information is duplicated across several applications.
- Typing long notes takes too much time.
- Handwritten notes and images cannot be searched easily.
- Users forget deadlines and unfinished tasks.
- Notes may be lost when a device is damaged or replaced.
- Some applications are too complicated for casual users.
- Users cannot always work without an Internet connection.



SEE

- Notes scattered across notebooks, messaging applications, photo galleries, and cloud storage.
- Screenshots without clear names or categories.
- Long lists of unfinished tasks.
- Different applications used for documents, calendars, drawings, and checklists.
- Important information hidden among hundreds of old notes.
- Users frequently switching between their phone and computer.

SAY & DO

- Write notes manually during lectures or meetings.
- Take photos of documents, whiteboards, receipts, and books.
- Send messages to themselves to save information.
- Create reminders using separate calendar applications.
- Search repeatedly through old files and conversations.
- Use voice input when they are walking, driving, or busy.
- Delete notes because they cannot find a better way to organize them.

GAIN

- Capture information quickly in any format.
- Find old notes within a few seconds.
- Keep study, work, and personal information organized.
- Receive reminders at the right time.
- Use the same notes on mobile and web.
- Feel confident that notes are automatically saved.
- Reduce the number of applications needed in daily life.
- Have a clean and comfortable interface for both beginners and experienced users.

User **persona**

Meet the everyday note-taker: **busy, visual, always on the move**

PERSONA 01



PRIMARY USER

Trần Minh Thư

Information

Age	21
Gender	Female
Job	3rd-year university student
Freelance	Freelance home tutor – Math & Informatics for 3 high-school students, 3 evenings/week; finds students via Facebook groups & parents' Zalo
Location	Danang, Vietnam
Marital status	Single
Income	~5.000.000 VND/month (~200 USD) from freelance tutoring + family support
Tech skill	●●●○○ **Mid-tech** – confident with everyday apps, cloud and Office; knows basic Markdown from CS courses but doesn't want to see syntax; can't write LaTeX by heart
Preferred devices	**Android phone** (primary – capture on the go, revising at night) · **Windows laptop** (lectures, long notes, preparing lessons)

Personality

Organized · Busy · Visual thinker · Practical

> *"My notes are everywhere: my phone, my laptop, a notebook, even Zalo. When I need one, I can never find it."*

Tools she uses today

Google Keep · Notion · Zalo "My Cloud" · Paper notebook · Phone camera

01 Bio

Thư is a third-year Computer Science student juggling lectures, group projects and evening tutoring.

She types notes in class, photographs the whiteboard, and jots ideas on the bus.

Before exams she revises on her phone late at night – and loses much of that time searching across apps. Three evenings a week she works as a freelance tutor; lesson plans, students' progress and tuition fees live in a paper notebook and a Zalo chat with parents.

02 Goals

- ✓ Capture an idea or a board photo in seconds.
- ✓ Keep lecture, project and tutoring notes in one place.
- ✓ Find any note by keyword when revising.
- ✓ Share notes with her group without pasting into chat.
- ✓ Run her tutoring like a small business: lesson plans, attendance and fees in one place.

03 Motivation

She wants to feel in control of a crowded week: less hunting for notes, more time to study and rest.

Look professional to students' parents – and never forget who has paid.

04 Needs

- ✓ One-tap capture for text, checklists and photos.
- ✓ Folders or tags per subject, with search inside photos.
- ✓ Works offline, syncs phone to laptop.
- ✓ Reminders attached to notes, next to the work.
- ✓ Write math formulas and tables for lessons without learning LaTeX.

05 Pain Points

- Notes scattered across five apps and a notebook.
- Workspace apps feel heavy for a quick note.
- Whiteboard photos get buried and can't be searched.
- Lost a semester of notes to a phone reset.
- Weak lecture-room Wi-Fi breaks cloud-only apps.
- Typing formulas on a phone is painful; she photographs handwritten ones instead.

07 Behavior

01 Types lecture notes on the laptop in class; photographs the whiteboard with her phone.

02 Jots ideas on the bus with one hand; uses voice when typing is awkward.

03 Prepares tutoring lessons on the laptop the night before; sends homework as photos via Zalo.

04 Tracks tuition fees and attendance in a paper notebook; checks Zalo to remember who paid.

05 Shares group-project notes by screenshot or copy-paste into Zalo.

06 Revises on her phone late at night in dark mode; checks the phone ~80 times a day.

07 Never backs up – relies on whatever the app does by default.

06 Summary

Thư needs a simple, unified note system that keeps her organized, searchable, and connected across devices – without the friction of switching apps, losing photos, or waiting for Wi-Fi.

Brand value

Google (Keep, Drive)	Free, familiar, syncs automatically
Zalo	Where her classmates, students and parents already are
Notion	Admires the clean look, but too heavy for quick notes
Canva	Beautiful results without design skills

free, simple, good-looking, reliable – a tool that "just works" and doesn't make her feel slow.

08 Opportunities

01 / **One-tap capture** from home screen widget + "+" button: text, checklist, photo, voice (1, 2, 4).

02 / **Scan whiteboard** with edge detection; **OCR text searchable** like normal text (3, 13).

03 / **Folder per subject**, **tag per student** `#tutoring/minh`; quick filter "Checklist chưa xong" before exams (7, 8, 15).

04 / **Tutoring templates**: Lesson plan, Attendance checklist, **Tuition table with auto** `=SUM` (6, 23.3).

05 / **Weekly repeating reminders** for each class; reminder on a single checklist item "Thu học phí Minh⏰" (16, 23.6).

06 / **LaTeX symbol palette + autocomplete** so she taps `\frac` instead of memorising it; Live Preview hides all syntax (23.4, 2.2).

07 / **Offline-first + phone ↔ laptop sync**, 30-day trash, version history – no more lost semester (19, C6).

08 / **Read-only link / PDF** to share lessons with students and notes with her group via Zalo (17, 18).

09 / **Dark mode + Focus mode** for late-night revision (20, 22).

User **persona**

Meet the plain-text perfectionist: **fast, structured, keyboard-first**

PERSONA 02



POWER USER

Phạm Đức Anh

Information

Age	27
Gender	Male
Job	**Backend developer at a software outsourcing company; part-time Master's in Data Science**
Location	**Danang, Vietnam**
Marital status	**Married (1 year), no children**
Income	~30,000,000 VND/month (~1,200 USD)
Tech skill	● ● ● ● ● **Expert** – writes Markdown & LaTeX daily, uses Git, lives on keyboard shortcuts, customises every tool
Preferred devices	**MacBook Pro** (primary – work & thesis) · **iPhone** (capture during commute & meetings) · company Windows PC

Personality

Analytical · Perfectionist · Introverted · Efficient

> *If I can't write it in Markdown and find it again in two seconds, it doesn't exist.*

Tools he uses today

Obsidian · VS Code · Notion (company wiki) · Apple Notes · Overleaf · GitHub

01 Bio

Đức Anh writes backend services by day and a Master's thesis on recommender systems by night.

His knowledge lives in Markdown: meeting notes, API docs, code snippets, paper summaries full of formulas.

He uses Obsidian on the Mac but finds its mobile app clunky and paid sync expensive, so ideas captured on the iPhone end up in Apple Notes and never get merged.

02 Goals

- ✓ Keep work, thesis and personal notes in **one plain-text knowledge base he owns**.
- ✓ Write formulas, tables and code without fighting the editor.
- ✓ Link ideas between papers, meetings and code (backlinks, embeds).
- ✓ Do everything from the keyboard; find any note in two seconds.
- ✓ Export clean `.md` / PDF to send to his thesis advisor.

03 Motivation

Build a "second brain" that will still open in ten years – and finish his thesis while working full time.

04 Needs

- ✓ Markdown-first editor with **Source mode**, Live Preview and Reading mode.
- ✓ **LaTeX autocomplete**, numbered equations, matrices.
- ✓ `[[backlinks]]`, link to headings/blocks, **embed notes**, aliases.
- ✓ Code blocks with syntax highlighting; tables that auto-align.
- ✓ **Query blocks** to list open tasks across notes.
- ✓ Plain `.md` export/import (Obsidian vault compatible); offline-first, fast free sync.
- ✓ Ctrl+K command palette, full keyboard shortcuts, dark mode.

05 Pain Points

- Obsidian sync is paid and its mobile app feels slow; notes split between Mac and iPhone.
- Notion is slow, uses proprietary blocks, and export breaks formulas.
- Equations in Word / Google Docs are painful; he rewrites them in Overleaf.
- Markdown tables are tedious to align by hand.
- Meeting tasks get lost inside long notes – no single "what's due" view.

07 Behavior

01. Opens the note app with a shortcut 30+ times a day; never touches the mouse while writing.
02. Writes meeting notes in Markdown with `- [ ]` tasks and `@date` deadlines.
03. Summarises 3–4 research papers a week with formulas and links between them.
04. Pastes code snippets and terminal output; tags by project `#work/api`, `#thesis/ch2`.
05. Captures quick ideas on the iPhone during the commute, refines them on the Mac.
06. Reviews a "dashboard" note every Monday listing open tasks.
07. Backs up everything in Git; distrusts tools that lock his data in.

06 Summary

Đức Anh needs a fast, Markdown-native note system that he fully owns – with first-class formulas, links and code – that syncs his Mac and iPhone for free and never locks his knowledge into a proprietary format.

Brand value	
Obsidian	Local Markdown files – he owns his data
GitHub	Version history, open formats, developer-first
Apple	Polish, speed, devices that work together
Linear / Raycast	Keyboard-first, instant, minimal

speed, open formats, privacy, keyboard control, no lock-in.

08 Opportunities

01 / **Source mode with line numbers**, Ctrl+E to switch Live Preview / Source / Reading (2.2).

02 / **LaTeX autocomplete** with Tab placeholders, `\tag{}`, `align`, matrices; KaTeX offline (23.4).

03 / **Backlinks, heading/block links, `![[embed]]`, aliases** to connect papers ↔ meetings ↔ code (11, 23.5).

04 / **Tables that auto-align**, paste from Excel/CSV, copy as LaTeX for the thesis (23.3).

05 / `- [ ] Fix API 📅 2026-01-12 @9:00` + **query block** dashboard of open tasks (16, 23.6, 23.7).

06 / **Code blocks** with highlighting, Mermaid diagrams for architecture (23.2, 23.8).

07 / **Ctrl+K palette, full shortcuts, slash menu** (13, 23.9).

08 / **Plain `.md` export/import**, Obsidian-vault compatible; line-level merge on sync conflict (19, 23.10).

09 / **Fast free sync Mac ↔ iPhone**, quick capture widget on iPhone (1, 19).

THANK YOU